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# Отчёт к лабораторной работе №4 по предмету «Администрирование в ОС Linux» «Работа с systemd»

## Часть 1

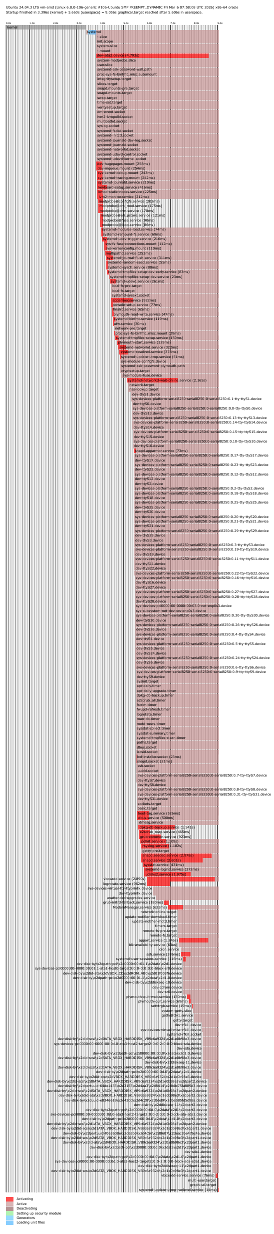
В данной части были использованы средства утилиты systemd-analyze для анализа времени загрузки системы. Команда systemd-analyze показала общее время загрузки ядра и пользовательского пространства. С помощью systemd-analyze blame был получен список сервисов, отсортированный по убыванию времени их инициализации. Команда systemd-analyze critical-chain sshd.service отобразила цепочку зависимостей, запуск которых предшествовал запуску сервиса SSH. Командой systemd-analyze plot был сформирован SVG-файл с графическим представлением процесса загрузки системы

```
[24-03-2026 01:20:36] {f6e2b08d} student@vm-amd:~$ systemd-analyze
Startup finished in 3.396s (kernel) + 5.660s (userspace) = 9.056s
graphical.target reached after 5.606s in userspace.
```

```
[24-03-2026 01:20:56] {f6e2b08d} student@vm-amd:~$ systemd-analyze blame
4.793s dev-sda2.device
2.978s snapd.seeded.service
2.890s vboxadd.service
2.601s snapd.service
2.165s systemd-networkd-wait-online.service
1.975s udisks2.service
1.541s dpkg-db-backup.service
1.246s apport.service
1.182s rsyslog.service
1.109s polkit.service
 962ms logrotate.service
 923ms grub-common.service
 922ms apparmor.service
 623ms ModemManager.service
 526ms boot-tag.service
 500ms dbus.service
 465ms e2scrub_reap.service
 431ms sysstat.service
 416ms keyboard-setup.service
 386ms ssh.service
 378ms systemd-resolved.service
 372ms systemd-logind.service
 322ms systemd-networkd.service
 311ms systemd-journal-flush.service
 261ms systemd-udevd.service
 258ms dev-hugepages.mount
 254ms dev-mqueue.mount
 253ms multipathd.service
 243ms sys-kernel-debug.mount
 242ms sys-kernel-tracing.mount
 225ms kmod-static-nodes.service
 216ms systemd-udev-trigger.service
 212ms lvm2-monitor.service
 210ms systemd-journald.service
 202ms modprobe@configfs.service
 196ms user@1000.service
 180ms grub-initrd-fallback.service
 175ms modprobe@dm_mod.service
 170ms modprobe@drm.service
 150ms systemd-tmpfiles-setup.service
 130ms plymouth-quit-wait.service
 129ms plymouth-start.service
 121ms modprobe@efi_pstore.service
 119ms systemd-binfmt.service
 116ms systemd-user-sessions.service
 112ms sys-fs-fuse-connections.mount
 110ms sys-kernel-config.mount
  90ms modprobe@fuse.service
  86ms modprobe@loop.service
  83ms systemd-tmpfiles-setup-dev-early.service
  80ms systemd-sysctl.service
  77ms console-setup.service
  76ms vboxadd-service.service
  74ms systemd-modules-load.service
  73ms snapd.apparmor.service
  69ms systemd-remount-fs.service
  69ms plymouth-quit.service
  65ms finalrd.service
  55ms systemd-random-seed.service
  51ms systemd-update-utmp.service
  47ms plymouth-read-write.service
  30ms ufw.service
```

```
  30ms ufw.service
  29ms proc-sys-fs-binfmt_misc.mount
  23ms lxd-installer.socket
  23ms systemd-tmpfiles-setup-dev.service
  21ms snapd.socket
  19ms setvtrgb.service
  17ms user-runtime-dir@1000.service
  14ms systemd-update-utmp-runlevel.service
   9ms sysstat-collect.service
   6ms e2scrub_all.service
  63us blk-availability.service
lines 11-72/72 (END)
```

```
[24-03-2026 01:24:27] {f6e2b08d} student@vm-amd:~$ systemd-analyze critical-chain sshd.service
The time when unit became active or started is printed after the "@" character.
The time the unit took to start is printed after the "+" character.

ssh.service +386ms
└─basic.target @2.158s
  └─sockets.target @2.158s
    └─snapd.socket @2.134s +21ms
      └─sysinit.target @2.116s
        └─snapd.apparmor.service @2.041s +73ms
          └─apparmor.service @1.083s +922ms
            └─local-fs.target @1.056s
              └─local-fs-pre.target @1.056s
                └─multipathd.service @802ms +253ms
                  └─systemd-remount-fs.service @673ms +69ms
                    └─systemd-journald.socket @441ms
                      └─.mount @429ms
                        └─.slice @428ms
[24-03-2026 01:24:35] {f6e2b08d} student@vm-amd:~$ systemd-analyze plot > boot_chart.svg
```



## Часть 2

В данной части были изучены основные команды управления юнитами systemd. С помощью systemctl list-units и list-unit-files получены списки запущенных сервисов и сервисов с автозагрузкой. Командой list-dependencies определены зависимости сервиса SSH. Проверен статус сервиса cron, а с помощью systemctl show выведены все его параметры, включая неявные. Командой systemctl disable отключена автозагрузка cron с сохранением возможности запуска по зависимостям

```
[[24-03-2026 01:25:14] {f6e2b08d} student@vm-amd:~$ systemctl list-units --type=service --state=running
  UNIT                         LOAD   ACTIVE SUB     DESCRIPTION
  cron.service                 loaded active running Regular background program processing daemon
  dbus.service                 loaded active running D-Bus System Message Bus
  getty@tty1.service           loaded active running Getty on tty1
  ModemManager.service         loaded active running Modem Manager
  multipathd.service           loaded active running Device-Mapper Multipath Device Controller
  polkit.service               loaded active running Authorization Manager
  rsyslog.service              loaded active running System Logging Service
  ssh.service                  loaded active running OpenBSD Secure Shell server
  systemd-journald.service     loaded active running Journal Service
  systemd-logind.service       loaded active running User Login Management
  systemd-networkd.service     loaded active running Network Configuration
  systemd-resolved.service     loaded active running Network Name Resolution
  systemd-udevd.service        loaded active running Rule-based Manager for Device Events and Files
  udisks2.service              loaded active running Disk Manager
  unattended-upgrades.service loaded active running Unattended Upgrades Shutdown
  user@1000.service            loaded active running User Manager for UID 1000

Legend: LOAD   → Reflects whether the unit definition was properly loaded.
        ACTIVE → The high-level unit activation state, i.e. generalization of SUB.
        SUB    → The low-level unit activation state, values depend on unit type.

16 loaded units listed.
```

```
[[24-03-2026 01:47:22] {f6e2b08d} student@vm-amd:~$ systemctl list-unit-files --type=service --state=enabled
UNIT FILE                            STATE   PRESET
apparmor.service                     enabled enabled
apport.service                       enabled enabled
blk-availability.service             enabled enabled
boot-tag.service                     enabled enabled
cloud-config.service                 enabled enabled
cloud-final.service                  enabled enabled
cloud-init-local.service             enabled enabled
cloud-init.service                   enabled enabled
console-setup.service                enabled enabled
cron.service                         enabled enabled
dmesg.service                        enabled enabled
e2scrub_reap.service                 enabled enabled
finalrd.service                      enabled enabled
getty@.service                       enabled enabled
gpu-manager.service                  enabled enabled
grub-common.service                  enabled enabled
grub-initrd-fallback.service         enabled enabled
keyboard-setup.service               enabled enabled
lvm2-monitor.service                 enabled enabled
ModemManager.service                 enabled enabled
multipathd.service                   enabled enabled
networkd-dispatcher.service          enabled enabled
open-iscsi.service                   enabled enabled
open-vm-tools.service                enabled enabled
pollinate.service                    enabled enabled
rsyslog.service                      enabled enabled
secureboot-db.service                enabled enabled
setvtrgb.service                     enabled enabled
snapd.apparmor.service               enabled enabled
snapd.autoimport.service             enabled enabled
snapd.core-fixup.service             enabled enabled
snapd.recovery-chooser-trigger.service enabled enabled
snapd.seeded.service                 enabled enabled
snapd.service                        enabled enabled
snapd.system-shutdown.service        enabled enabled
ssh.service                          enabled enabled
sysstat.service                      enabled enabled
systemd-networkd-wait-online.service enabled enabled
systemd-networkd.service             enabled enabled
systemd-pstore.service               enabled enabled
systemd-resolved.service             enabled enabled
systemd-timesyncd.service            enabled enabled
thermald.service                     enabled enabled
ua-reboot-cmds.service               enabled enabled
ubuntu-advantage.service             enabled enabled
udisks2.service                      enabled enabled
ufw.service                          enabled enabled
unattended-upgrades.service          enabled enabled
vboxadd-service.service              enabled enabled
vboxadd.service                      enabled enabled
vgauth.service                       enabled enabled

51 unit files listed.
```

```
[[24-03-2026 01:47:33] {f6e2b08d} student@vm-amd:~$ systemctl list-dependencies ssh.service
ssh.service
● ├─-.mount
● ├─ssh.socket
● ├─system.slice
● └─sysinit.target
●   ├─apparmor.service
●   ├─blk-availability.service
●   ├─dev-hugepages.mount
●   ├─dev-mqueue.mount
●   ├─finalrd.service
●   ├─keyboard-setup.service
●   ├─kmod-static-nodes.service
○   ├─ldconfig.service
●   ├─lvm2-lvmpolld.socket
●   ├─lvm2-monitor.service
●   ├─multipathd.service
○   ├─open-iscsi.service
●   ├─plymouth-read-write.service
●   ├─plymouth-start.service
●   ├─proc-sys-fs-binfmt_misc.automount
●   ├─setvtrgb.service
●   ├─sys-fs-fuse-connections.mount
●   ├─sys-kernel-config.mount
●   ├─sys-kernel-debug.mount
●   ├─sys-kernel-tracing.mount
○   ├─systemd-ask-password-console.path
●   ├─systemd-binfmt.service
○   ├─systemd-firstboot.service
○   ├─systemd-hwdb-update.service
○   ├─systemd-journal-catalog-update.service
●   ├─systemd-journal-flush.service
●   ├─systemd-journald.service
○   ├─systemd-machine-id-commit.service
●   ├─systemd-modules-load.service
○   ├─systemd-pcrmachine.service
○   ├─systemd-pcrphase-sysinit.service
○   ├─systemd-pcrphase.service
○   ├─systemd-pstore.service
●   ├─systemd-random-seed.service
○   ├─systemd-repart.service
●   ├─systemd-resolved.service
●   ├─systemd-sysctl.service
○   ├─systemd-sysusers.service
○   ├─systemd-timesyncd.service
●   ├─systemd-tmpfiles-setup-dev-early.service
●   ├─systemd-tmpfiles-setup-dev.service
●   ├─systemd-tmpfiles-setup.service
○   ├─systemd-tpm2-setup-early.service
○   ├─systemd-tpm2-setup.service
●   ├─systemd-udev-trigger.service
●   ├─systemd-udevd.service
○   ├─systemd-update-done.service
●   ├─systemd-update-utmp.service
●   ├─cryptsetup.target
●   ├─integritysetup.target
●   ├─local-fs.target
●   │ ├─-.mount
○   │ ├─systemd-fsck-root.service
●   │ └─systemd-remount-fs.service
●   ├─swap.target
●   └─veritysetup.target
```

```
[[24-03-2026 01:48:33] {f6e2b08d} student@vm-amd:~$ systemctl status cron.service
● cron.service - Regular background program processing daemon
     Loaded: loaded (/usr/lib/systemd/system/cron.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-03-24 01:18:44 MSK; 30min ago
       Docs: man:cron(8)
   Main PID: 759 (cron)
      Tasks: 1 (limit: 2265)
     Memory: 440.0K (peak: 1.9M)
        CPU: 56ms
     CGroup: /system.slice/cron.service
             └─759 /usr/sbin/cron -f -P

Warning: some journal files were not opened due to insufficient permissions.
```

```
[[24-03-2026 01:49:26] {f6e2b08d} student@vm-amd:~$ systemctl show cron.service
Type=simple
ExitType=main
Restart=on-failure
RestartMode=normal
NotifyAccess=none
RestartUSec=100ms
RestartSteps=0
RestartMaxDelayUSec=infinity
RestartUSecNext=100ms
TimeoutStartUSec=1min 30s
TimeoutStopUSec=1min 30s
TimeoutAbortUSec=1min 30s
TimeoutStartFailureMode=terminate
TimeoutStopFailureMode=terminate
RuntimeMaxUSec=infinity
RuntimeRandomizedExtraUSec=0
WatchdogUSec=0
WatchdogTimestampMonotonic=0
RootDirectoryStartOnly=no
RemainAfterExit=no
GuessMainPID=yes
MainPID=759
ControlPID=0
FileDescriptorStoreMax=0
NFileDescriptorStore=0
FileDescriptorStorePreserve=restart
StatusErrno=0
Result=success
ReloadResult=success
CleanResult=success
UID=[not set]
GID=[not set]
NRestarts=0
OOMPolicy=stop
ReloadSignal=1
ExecMainStartTimestamp=Tue 2026-03-24 01:18:44 MSK
ExecMainStartTimestampMonotonic=7423757
ExecMainExitTimestampMonotonic=0
ExecMainPID=759
ExecMainCode=0
ExecMainStatus=0
ExecStart={ path=/usr/sbin/cron ; argv[]=/usr/sbin/cron -f -P $EXTRA_OPTS ; ignore_errors=no ; start_time=[Tue >
ExecStartEx={ path=/usr/sbin/cron ; argv[]=/usr/sbin/cron -f -P $EXTRA_OPTS ; flags= ; start_time=[Tue 2026-03->
Slice=system.slice
ControlGroup=/system.slice/cron.service
ControlGroupId=2818
MemoryCurrent=450560
MemoryPeak=2031616
MemorySwapCurrent=0
MemorySwapPeak=0
MemoryZSwapCurrent=0
MemoryAvailable=1737986048
```

```
[[24-03-2026 01:50:40] {f6e2b08d} student@vm-amd:~$ sudo systemctl disable cron.service
[sudo] password for student:
Synchronizing state of cron.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install disable cron
Removed "/etc/systemd/system/multi-user.target.wants/cron.service".
```

## Часть 3

Создан пользовательский сервис mymsg, который при запуске записывает в системный журнал текущую дату и время с помощью команды logger. Сервис настроен с типом oneshot и зависимостью от network.target, что гарантирует его запуск только после инициализации сети. Корректность юнит-файла проверена утилитой systemd-analyze verify. Настроена автозагрузка сервиса и выполнен его ручной запуск

```
[24-03-2026 01:51:25] {f6e2b08d} student@vm-amd:~$ sudo nano /etc/systemd/system/mymsg.service
[24-03-2026 01:55:34] {f6e2b08d} student@vm-amd:~$ systemd-analyze verify /etc/systemd/system/mymsg.service
[24-03-2026 01:55:44] {f6e2b08d} student@vm-amd:~$ sudo systemctl daemon-reload
[24-03-2026 01:55:52] {f6e2b08d} student@vm-amd:~$ sudo systemctl enable mymsg.service
Created symlink /etc/systemd/system/multi-user.target.wants/mymsg.service → /etc/systemd/system/mymsg.service.
[24-03-2026 01:56:05] {f6e2b08d} student@vm-amd:~$ sudo systemctl start mymsg.service
[24-03-2026 01:56:11] {f6e2b08d} student@vm-amd:~$ systemctl status mymsg.service
● mymsg.service - My Message Service
     Loaded: loaded (/etc/systemd/system/mymsg.service; enabled; preset: enabled)
     Active: active (exited) since Tue 2026-03-24 01:56:11 MSK; 5s ago
    Process: 1949 ExecStart=/usr/bin/logger mymsg: System started at $(date) (code=exited, status=0/SUCCESS)
   Main PID: 1949 (code=exited, status=0/SUCCESS)
        CPU: 5ms
```

```
  GNU nano 7.2                              /etc/systemd/system/mymsg.service *
[Unit]
Description=My Message Service
After=network.target
Requires=network.target

[Service]
Type=oneshot
ExecStart=/usr/bin/logger "mymsg: System started at $(date)"
RemainAfterExit=yes

[Install]
WantedBy=multi-user.target
```

## Часть 4

В данной части была изучена работа с системным журналом systemd через утилиту journalctl. Проверена корректность работы сервиса mymsg - в журнале присутствуют записи о его запуске и завершении. С помощью фильтра -u выведены сообщения конкретного сервиса, а с помощью -p err - все сообщения с уровнем ошибок. Командой --disk-usage определён размер журнала - 87.0 МБ

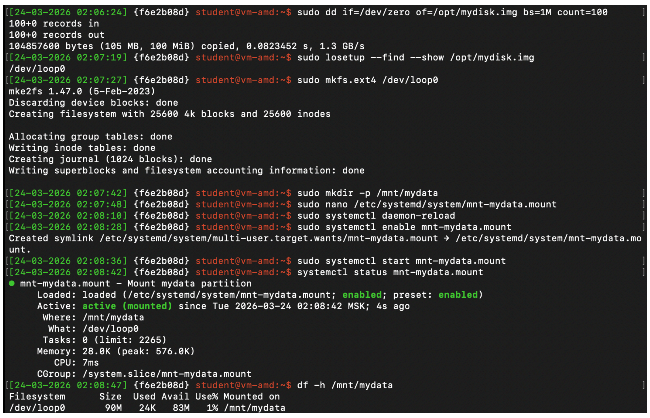
```
[24-03-2026 02:02:21] {f6e2b08d} student@vm-amd:~$ sudo journalctl -e
Mar 24 01:51:25 vm-amd dbus-daemon[563]: [system] Activating via systemd: service name='org.freedesktop.UPower'>
Mar 24 01:51:25 vm-amd systemd[1]: Starting upower.service - Daemon for power management...
Mar 24 01:51:26 vm-amd dbus-daemon[563]: [system] Successfully activated service 'org.freedesktop.UPower'
Mar 24 01:51:26 vm-amd systemd[1]: Started upower.service - Daemon for power management.
Mar 24 01:51:26 vm-amd fwupd[1490]: 22:51:26.175 FuEngine             failed to add device /sys/devices/pci0000>
Mar 24 01:51:26 vm-amd fwupd[1490]: 22:51:26.209 FuMain               fwupd 1.9.30 ready for requests (locale e>
Mar 24 01:51:26 vm-amd dbus-daemon[563]: [system] Successfully activated service 'org.freedesktop.fwupd'
Mar 24 01:51:26 vm-amd systemd[1]: Started fwupd.service - Firmware update daemon.
Mar 24 01:51:26 vm-amd fwupdmgr[1469]: Updating lvfs
Mar 24 01:51:28 vm-amd fwupdmgr[1469]: Successfully downloaded new metadata: 0 local devices supported
Mar 24 01:51:28 vm-amd systemd[1]: fwupd-refresh.service: Deactivated successfully.
Mar 24 01:51:28 vm-amd systemd[1]: Finished fwupd-refresh.service - Refresh fwupd metadata and update motd.
Mar 24 01:51:35 vm-amd apt.systemd.daily[1517]: /usr/bin/unattended-upgrade:567: DeprecationWarning: This proce>
Mar 24 01:51:35 vm-amd apt.systemd.daily[1517]:   pid = os.fork()
Mar 24 01:51:43 vm-amd dbus-daemon[563]: [system] Activating via systemd: service name='org.freedesktop.Package>
Mar 24 01:51:43 vm-amd systemd[1]: Starting packagekit.service - PackageKit Daemon...
Mar 24 01:51:43 vm-amd PackageKit[1752]: daemon start
Mar 24 01:51:43 vm-amd dbus-daemon[563]: [system] Successfully activated service 'org.freedesktop.PackageKit'
Mar 24 01:51:43 vm-amd systemd[1]: Started packagekit.service - PackageKit Daemon.
Mar 24 01:51:47 vm-amd systemd[1]: apt-daily-upgrade.service: Deactivated successfully.
Mar 24 01:51:47 vm-amd systemd[1]: Finished apt-daily-upgrade.service - Daily apt upgrade and clean activities.
Mar 24 01:51:47 vm-amd systemd[1]: apt-daily-upgrade.service: Consumed 15.654s CPU time, 2.2M memory peak, 0B m>
Mar 24 01:54:49 vm-amd sudo[1847]:  student : TTY=pts/0 ; PWD=/home/student ; USER=root ; COMMAND=/usr/bin/nano>
Mar 24 01:54:49 vm-amd sudo[1847]: pam_unix(sudo:session): session opened for user root(uid=0) by student(uid=1>
Mar 24 01:55:01 vm-amd CRON[1851]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Mar 24 01:55:01 vm-amd CRON[1852]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
Mar 24 01:55:01 vm-amd CRON[1851]: pam_unix(cron:session): session closed for user root
Mar 24 01:55:34 vm-amd sudo[1847]: pam_unix(sudo:session): session closed for user root
Mar 24 01:55:51 vm-amd sudo[1857]:  student : TTY=pts/0 ; PWD=/home/student ; USER=root ; COMMAND=/usr/bin/syst>
Mar 24 01:55:51 vm-amd sudo[1857]: pam_unix(sudo:session): session opened for user root(uid=0) by student(uid=1>
Mar 24 01:55:51 vm-amd systemd[1]: Reloading requested from client PID 1859 ('systemctl') (unit session-1.scope>
Mar 24 01:55:51 vm-amd systemd[1]: Reloading...
Mar 24 01:55:52 vm-amd systemd[1]: Reloading finished in 285 ms.
Mar 24 01:55:52 vm-amd sudo[1857]: pam_unix(sudo:session): session closed for user root
Mar 24 01:56:04 vm-amd sudo[1901]:  student : TTY=pts/0 ; PWD=/home/student ; USER=root ; COMMAND=/usr/bin/syst>
Mar 24 01:56:04 vm-amd sudo[1901]: pam_unix(sudo:session): session opened for user root(uid=0) by student(uid=1>
Mar 24 01:56:04 vm-amd systemd[1]: Reloading requested from client PID 1903 ('systemctl') (unit session-1.scope>
Mar 24 01:56:04 vm-amd systemd[1]: Reloading...
Mar 24 01:56:05 vm-amd systemd[1]: Reloading finished in 293 ms.
Mar 24 01:56:05 vm-amd sudo[1901]: pam_unix(sudo:session): session closed for user root
Mar 24 01:56:11 vm-amd sudo[1945]:  student : TTY=pts/0 ; PWD=/home/student ; USER=root ; COMMAND=/usr/bin/syst>
Mar 24 01:56:11 vm-amd sudo[1945]: pam_unix(sudo:session): session opened for user root(uid=0) by student(uid=1>
Mar 24 01:56:11 vm-amd systemd[1]: Starting mymsg.service - My Message Service...
```

```
lines 1-54/54 (END)
[24-03-2026 02:02:20] {f6e2b08d} student@vm-amd:~$ sudo journalctl --disk-usage
Archived and active journals take up 87.0M in the file system.
```

```
[24-03-2026 02:00:09] {f6e2b08d} student@vm-amd:~$ sudo journalctl -u mymsg.service
Mar 24 01:56:11 vm-amd systemd[1]: Starting mymsg.service - My Message Service...
Mar 24 01:56:11 vm-amd systemd[1]: Finished mymsg.service - My Message Service.
[24-03-2026 02:01:57] {f6e2b08d} student@vm-amd:~$ sudo journalctl -p err
Feb 16 22:59:27 vm-amd kernel: RETBleed: WARNING: Spectre v2 mitigation leaves CPU vulnerable to RETBleed attac>
Feb 16 22:59:27 vm-amd kernel: ITS: WARNING: ITS mitigation depends on retpoline and rethunk support
Feb 16 22:59:32 vm-amd kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an unsupported >
Feb 16 22:59:32 vm-amd kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely broken.
Feb 16 22:59:32 vm-amd kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported graphics device >
Feb 16 22:59:33 vm-amd systemd[1]: Failed to start vboxadd-service.service.
-- Boot 7dfe55310c9a40aaa83357006ecbbce1 --
Feb 16 23:02:14 vm-amd kernel: RETBleed: WARNING: Spectre v2 mitigation leaves CPU vulnerable to RETBleed attac>
Feb 16 23:02:14 vm-amd kernel: ITS: WARNING: ITS mitigation depends on retpoline and rethunk support
Feb 16 23:02:18 vm-amd kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an unsupported >
Feb 16 23:02:18 vm-amd kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely broken.
Feb 16 23:02:18 vm-amd kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported graphics device >
Feb 16 23:02:18 vm-amd systemd[1]: Failed to start vboxadd-service.service.
-- Boot 4bc342ca4278467ba1e4eda961ff15d9 --
Feb 16 23:05:58 vm-amd kernel: RETBleed: WARNING: Spectre v2 mitigation leaves CPU vulnerable to RETBleed attac>
Feb 16 23:05:58 vm-amd kernel: ITS: WARNING: ITS mitigation depends on retpoline and rethunk support
Feb 16 23:06:04 vm-amd kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an unsupported >
Feb 16 23:06:04 vm-amd kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely broken.
Feb 16 23:06:04 vm-amd kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported graphics device >
Feb 16 23:06:04 vm-amd systemd[1]: Failed to start vboxadd-service.service.
-- Boot b30c6f6ece80471ca84938abee6c1099 --
Feb 16 23:49:45 vm-amd kernel: RETBleed: WARNING: Spectre v2 mitigation leaves CPU vulnerable to RETBleed attac>
Feb 16 23:49:45 vm-amd kernel: ITS: WARNING: ITS mitigation depends on retpoline and rethunk support
Feb 16 23:49:50 vm-amd kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an unsupported >
Feb 16 23:49:50 vm-amd kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely broken.
Feb 16 23:49:50 vm-amd kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported graphics device >
Feb 16 23:49:50 vm-amd systemd[1]: Failed to start vboxadd-service.service.
Feb 16 23:52:24 vm-amd (systemd)[985]: PAM failed: Authentication token is no longer valid; new one required
Feb 16 23:52:24 vm-amd (systemd)[985]: user@1000.service: Failed to set up PAM session: Operation not permitted
Feb 16 23:52:24 vm-amd systemd[1]: Failed to start user@1000.service - User Manager for UID 1000.
Feb 17 01:11:17 vm-amd sudo[2703]:       u1 : command not allowed ; TTY=pts/1 ; PWD=/home/student/lab1 ; USER=r>
Feb 17 01:35:31 vm-amd sudo[2950]:       u1 : command not allowed ; TTY=pts/1 ; PWD=/home/student/lab1 ; USER=r>
-- Boot a5e3535f11254e2cb2a42646cdef0cc0 --
Feb 17 13:41:12 vm-amd kernel: RETBleed: WARNING: Spectre v2 mitigation leaves CPU vulnerable to RETBleed attac>
Feb 17 13:41:12 vm-amd kernel: ITS: WARNING: ITS mitigation depends on retpoline and rethunk support
Feb 17 13:41:18 vm-amd kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an unsupported >
Feb 17 13:41:18 vm-amd kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely broken.
Feb 17 13:41:18 vm-amd kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported graphics device >
Feb 17 13:41:19 vm-amd systemd[1]: Failed to start vboxadd-service.service.
-- Boot 6e80a8fd04e04589836105d71c74b069 --
Mar 16 18:03:44 vm-amd kernel: RETBleed: WARNING: Spectre v2 mitigation leaves CPU vulnerable to RETBleed attac>
Mar 16 18:03:44 vm-amd kernel: ITS: WARNING: ITS mitigation depends on retpoline and rethunk support
Mar 16 18:03:46 vm-amd systemd[1]: Failed to start rsyslog.service - System Logging Service.
```

**Часть 5**

В данной части создан и настроен .mount юнит systemd для монтирования файловой системы. Поскольку на диске отсутствовали свободные разделы, был создан файл-образ диска размером 100 МБ, подключённый через loop-устройство и отформатированный в ext4. Создан юнит-файл mnt-mydata.mount с указанием устройства, точки монтирования и типа ФС. Юнит включён в автозагрузку и успешно запущен - монтирование подтверждено командами systemctl status и df -h

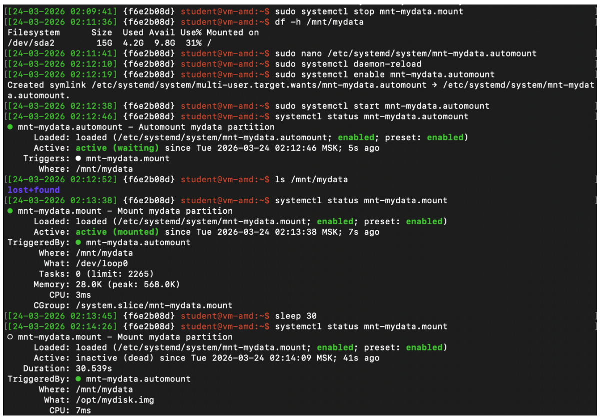


```
[[24-03-2026 02:06:24] {f6e2b08d} student@vm-amd:~$ sudo dd if=/dev/zero of=/opt/mydisk.img bs=1M count=100
100+0 records in
100+0 records out
104857600 bytes (105 MB, 100 MiB) copied, 0.0823452 s, 1.3 GB/s
[[24-03-2026 02:07:19] {f6e2b08d} student@vm-amd:~$ sudo losetup --find --show /opt/mydisk.img
/dev/loop0
[[24-03-2026 02:07:27] {f6e2b08d} student@vm-amd:~$ sudo mkfs.ext4 /dev/loop0
mke2fs 1.47.0 (5-Feb-2023)
Discarding device blocks: done
Creating filesystem with 25600 4k blocks and 25600 inodes

Allocating group tables: done
Writing inode tables: done
Creating journal (1024 blocks): done
Writing superblocks and filesystem accounting information: done

[[24-03-2026 02:07:42] {f6e2b08d} student@vm-amd:~$ sudo mkdir -p /mnt/mydata
[[24-03-2026 02:07:48] {f6e2b08d} student@vm-amd:~$ sudo nano /etc/systemd/system/mnt-mydata.mount
[[24-03-2026 02:08:10] {f6e2b08d} student@vm-amd:~$ sudo systemctl daemon-reload
[[24-03-2026 02:08:28] {f6e2b08d} student@vm-amd:~$ sudo systemctl enable mnt-mydata.mount
Created symlink /etc/systemd/system/multi-user.target.wants/mnt-mydata.mount → /etc/systemd/system/mnt-mydata.mo
unt.
[[24-03-2026 02:08:36] {f6e2b08d} student@vm-amd:~$ sudo systemctl start mnt-mydata.mount
[[24-03-2026 02:08:42] {f6e2b08d} student@vm-amd:~$ systemctl status mnt-mydata.mount
● mnt-mydata.mount - Mount mydata partition
     Loaded: loaded (/etc/systemd/system/mnt-mydata.mount; enabled; preset: enabled)
     Active: active (mounted) since Tue 2026-03-24 02:08:42 MSK; 4s ago
      Where: /mnt/mydata
       What: /dev/loop0
      Tasks: 0 (limit: 2265)
     Memory: 28.0K (peak: 576.0K)
        CPU: 7ms
     CGroup: /system.slice/mnt-mydata.mount
[[24-03-2026 02:08:47] {f6e2b08d} student@vm-amd:~$ df -h /mnt/mydata
Filesystem      Size  Used Avail Use% Mounted on
/dev/loop0       90M   24K   83M   1% /mnt/mydata
```

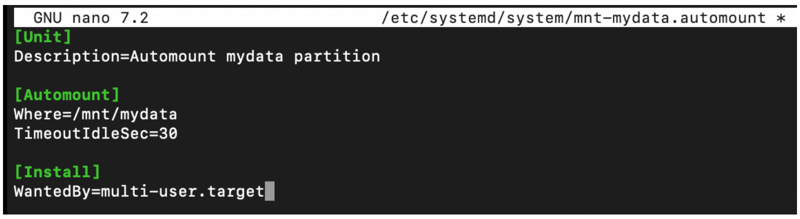
```
  GNU nano 7.2                     /etc/systemd/system/mnt-mydata.mount
[Unit]
Description=Mount mydata partition

[Mount]
What=/opt/mydisk.img
Where=/mnt/mydata
Type=ext4
Options=loop,defaults

[Install]
WantedBy=multi-user.target
```

**Часть 6**

Создан .automount юнит для реализации отложенного монтирования. При обращении к точке монтирования /mnt/mydata раздел автоматически монтируется, а после 30 секунд бездействия - автоматически размонтируется. Это позволяет экономить системные ресурсы, монтируя раздел только по требованию



```
[[24-03-2026 02:09:41] {f6e2b08d} student@vm-amd:~$ sudo systemctl stop mnt-mydata.mount
[[24-03-2026 02:11:36] {f6e2b08d} student@vm-amd:~$ df -h /mnt/mydata
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda2        15G  4.2G  9.8G  31% /
[[24-03-2026 02:11:41] {f6e2b08d} student@vm-amd:~$ sudo nano /etc/systemd/system/mnt-mydata.automount
[[24-03-2026 02:12:10] {f6e2b08d} student@vm-amd:~$ sudo systemctl daemon-reload
[[24-03-2026 02:12:19] {f6e2b08d} student@vm-amd:~$ sudo systemctl enable mnt-mydata.automount
Created symlink /etc/systemd/system/multi-user.target.wants/mnt-mydata.automount → /etc/systemd/system/mnt-mydat
a.automount.
[[24-03-2026 02:12:38] {f6e2b08d} student@vm-amd:~$ sudo systemctl start mnt-mydata.automount
[[24-03-2026 02:12:46] {f6e2b08d} student@vm-amd:~$ systemctl status mnt-mydata.automount
● mnt-mydata.automount - Automount mydata partition
     Loaded: loaded (/etc/systemd/system/mnt-mydata.automount; enabled; preset: enabled)
     Active: active (waiting) since Tue 2026-03-24 02:12:46 MSK; 5s ago
   Triggers: ● mnt-mydata.mount
      Where: /mnt/mydata
[[24-03-2026 02:12:52] {f6e2b08d} student@vm-amd:~$ ls /mnt/mydata
lost+found
[[24-03-2026 02:13:38] {f6e2b08d} student@vm-amd:~$ systemctl status mnt-mydata.mount
● mnt-mydata.mount - Mount mydata partition
     Loaded: loaded (/etc/systemd/system/mnt-mydata.mount; enabled; preset: enabled)
     Active: active (mounted) since Tue 2026-03-24 02:13:38 MSK; 7s ago
TriggeredBy: ● mnt-mydata.automount
      Where: /mnt/mydata
       What: /dev/loop0
      Tasks: 0 (limit: 2265)
     Memory: 28.0K (peak: 568.0K)
        CPU: 3ms
     CGroup: /system.slice/mnt-mydata.mount
[[24-03-2026 02:13:45] {f6e2b08d} student@vm-amd:~$ sleep 30
[[24-03-2026 02:14:26] {f6e2b08d} student@vm-amd:~$ systemctl status mnt-mydata.mount
○ mnt-mydata.mount - Mount mydata partition
     Loaded: loaded (/etc/systemd/system/mnt-mydata.mount; enabled; preset: enabled)
     Active: inactive (dead) since Tue 2026-03-24 02:14:09 MSK; 41s ago
   Duration: 30.539s
TriggeredBy: ● mnt-mydata.automount
      Where: /mnt/mydata
       What: /opt/mydisk.img
        CPU: 7ms
```



```
  GNU nano 7.2                     /etc/systemd/system/mnt-mydata.automount *
[Unit]
Description=Automount mydata partition

[Automount]
Where=/mnt/mydata
TimeoutIdleSec=30

[Install]
WantedBy=multi-user.target
```

**Вопросы:**

1) systemctl restart останавливает и запускает сервис безусловно, даже если он не был запущен, он будет запущен. systemctl try-restart перезапускает сервис только в том случае, если он уже запущен, если сервис не активен, команда ничего не сделает

2) sudo systemctl isolate rescue.target
3)
В юните mysrv.service:
Секция [Unit]: Conflicts=mymsg.service - mysrv и mymsg не могут работать одновременно, при запуске mysrv сервис mymsg будет остановлен, и наоборот
Секция [Install]: отсутствует WantedBy, поэтому сервис не будет запускаться автоматически

В юните mymsg.service:
Секция [Unit]: Conflicts=mysrv.service - зеркальное указание конфликта
Таким образом, mysrv можно запустить только вручную и только когда mymsg остановлен, либо mymsg автоматически остановится при запуске mysrv из-за Conflicts